

LESSON 2-3 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2

Finding the Median

Maryland Standard 6.SP.B.5c · Learning target – I can find the median of a data set by ordering the values and taking the middle one.

REMEMBER

- The **median** is the middle value of a data set.
- First **order** the values from least to greatest.
- With an odd number of values, one value sits in the middle.

WORKED EXAMPLE

Find the median of 14, 9, 21, 6, 17.

Step 1 Order: 6, 9, 14, 17, 21

Step 2 Cross off one from each end until one is left:
14.

Answer Median = 14

WORD BANK

median – the middle number

data set – a list of numbers

order – put numbers from least to greatest

PRACTICE

show your work · label units

1. Find the median of each data set.

a. 8, 3, 6 _____

b. 12, 18, 10, 15, 11 _____

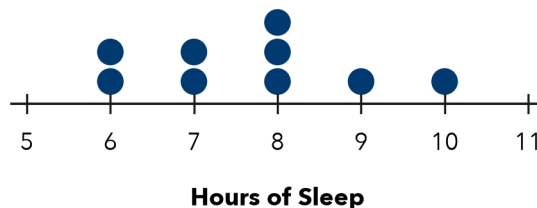
c. 25, 40, 35, 30, 20 _____

d. 7, 2, 9, 4, 5, 8, 1 _____

2. **Sleep.** The dot plot shows how many hours some students slept last night.

a. Number of students: _____

b. Median: _____ hours



3. **Bike Rides.** Kiara rode her bike 7 times. She rode 4, 11, 7, 15, 9, 6, and 12 miles. Find the median distance.

Step 1 Write the 7 distances in order.

Step 2 Cross off one from each end until one is left.

Show your work.

Answer _____ miles

4. **Missing Score.** Four of five scores are **82, 90, 75, 88**. The **median** is **85**. Find the fifth score.

Step 1 Put the four scores in order.

Step 2 The median is the 3rd of 5 values. Where can 85 fit?

Answer _____ points

5. **Build a Set.** Write a set of 5 different whole numbers that has a median of 12.

Step 1 Put 12 in the middle.

Step 2 Write two numbers less than 12 and two numbers greater than 12.

My set _____

6. **Reason.** The data set 4, 6, 9, 11, 30 has a median of 9. If 30 changes to 100, does the median change? Explain.

Frame The median ___ change because ___.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Ask 5 people at home how many letters are in their first name. Find the median.

LESSON 2-3 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Median of an Even Data Set**

Maryland Standard 6.SP.B.5c • Learning target – I can find the median of a data set with an even number of values by averaging the two middle values.

REMEMBER

- With an **even** number of values, there are **two middle values**.
- The median is halfway between them: **add the two, then divide by 2**.
- The median does not have to be a data value.

WORKED EXAMPLE

Find the median of 12, 7, 15, 10, 20, 9.

Step 1 Order: 7, 9, 10, 12, 15, 20

Step 2 Two middle values: 10 and 12

Step 3 $(10 + 12) \div 2 = 22 \div 2 = 11$

Answer Median = 11

WORD BANK

median – the middle number

average – add, then divide by 2

middle values – the two numbers in the middle

PRACTICE

show your work • label units

1. Find the median of each data set.

Step 1 Order the numbers.

Step 2 Find the two middle numbers.

Step 3 Add them and divide by 2.

a. 3, 9, 5, 7 _____

b. 20, 14, 18, 12 _____

c. 11, 6, 9, 4, 13, 8 _____

d. 30, 45, 35, 50, 40, 25 _____

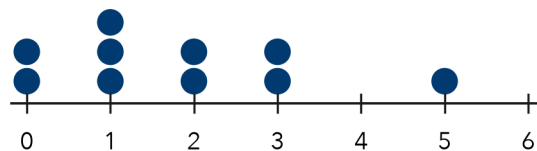
e. 16, 9, 21, 12, 30, 14, 25, 10 _____

f. 5, 11, 8, 5 _____

2. **Cousins.** The dot plot shows how many cousins 10 students have.

a. Two middle values: _____

b. Median: _____



Number of Cousins

3. **Puppies.** Six puppies weigh 8, 12, 5, 15, 10, and 7 pounds. Find the median weight.

Step 1 Order the six weights.

Step 2 Average the two middle weights.

Answer _____ pounds

4. **Work Backward.** Four values in order are 6, 10, ____, 20. The median is 12. Find the missing value.

Step 1 The median is 12, so the two middle numbers add to 12×2 .

Step 2 Subtract 10 from that sum.

Missing value _____

5. **Find the Error.** Ana says the median of 8, 2, 10, 6 is $(2 + 10) \div 2 = 6$. Explain her error.

Frame Ana forgot to ____. The correct median is ____.

TEST PRACTICE

choose the best answer

6. What is the median of 15, 9, 12, 20?

A 13.5

B 12

C 14

D 10.5

7. Six values have a median of 30. Which could be the two middle values?

A 28 and 32

B 30 and 34

C 25 and 30

D 29 and 30

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Ask 4 or 6 people at home their shoe size. Find the median.